

Anexa 3.1 - CV and track record template (max. 4 pages)

CURRICULUM VITAE

Personal information

Name, Surname:	Fran Nekvapil		
Date of birth:	16. August 1993	Gender:	M
Nationality:	Croatian		
Researcher unique identifier(s) (ORCID, Researcher ID etc.):	UEF-ID (BrainMap): U-1900-061G-9020 ORCID: https://orcid.org/0000-0001-8652-8292		
Personal link from the platform at https://www.brainmap.ro/	https://www.brainmap.ro/fran-nekvapil		
URL for personal website (if case):	https://en.itim-cj.ro/portfolio/nekvapil-fran/		

Education **POCU?**

Year	Faculty/department - University/institution - Country
2020 (dissertation defended)	Ph.D. in Physics; Babeş-Bolyai University, Faculty of Physics; Romania
2017	Master degree “Mariculture”; University of Dubrovnik, Department of Aquaculture; Croatia

Positions - current and previous

(Academic sector/research institutes/industrial sector/public sector/other)

Year	Job title – Employer - Country
2018 - present	R1 First stage researcher (Dec. 2025) – full-time, permanent employment National Institute for Research and Development of Isotopic and Molecular Technologies; research institute; Romania
2020 – present	R2 Recognized researcher (Jul. 2025) – part-time, employment only on projects Babeş-Bolyai University, Ioan Ursu Institute; higher education institution; Romania

Career breaks (if case)

Year	Reason
YYYY-YYYY	

Project management experience

(Academic sector/research institutes/industrial sector/public sector/other. Please list the most relevant.)

Year	Project title - Role – Funder – Budget – link to project webpage
2022 - 2023	New composites based on carotenoid-depleted seafood shell waste with efficient environmental pollutants adsorption; Project Leader (PD-type project); funder: UEFISCDI; budget 218 000 RON (cca. 43 500 EUR), 18 months; https://shellpolads.granturi.ubbcluj.ro/
2020 - 2021	Blue bioeconomy approach to wasted marine biomass - an innovative fertilizer for green economy and agriculture (acronym); Project Leader (internal BBU competition); funder Babeş-Bolyai University; budget 50 000 RON (cca. 10 000 EUR), 15 months

Other relevant professional experiences

(e.g. institutional responsibilities, organisation of scientific meetings, membership in academic societies, review boards, advisory boards, committees and major research or innovation collaborations, other commissions of trust in public or private sector)

Year	Description - Role
2023	ERASMUS+ Staff exchange at Chemnitz University of Technology, Germany (24.6.-8.7.2023), with focus on Raman spectroscopy techniques applied to metallic nanoparticles
2021	Main organizer First workshop on Blue Bioeconomy approach to Marine Biomass Waste: Innovative Ideas, Techniques and Translational Sciences <i>With participation of the company Metrohm Romania, branch B&W TEK at Babeş-Bolyai University, Cluj-Napoca (Romania)</i> https://bluebiosustain.granturi.ubbcluj.ro/dissem.html#Workshops
2021	Main organizer First workshop on Blue Bioeconomy approach to Marine Biomass Waste: Innovative Ideas, Techniques and Translational Sciences <i>Reinforcing the collaboration between the Babeş-Bolyai University and the University of Dubrovnik</i> At University of Dubrovnik, Dubrovnik (Croatia) https://bluebiosustain.granturi.ubbcluj.ro/dissem.html#Workshops
2017	ERASMUS+ placement (9 months) at Raman-AFM lab, Faculty of Physics, Babeş-Bolyai University
2016	ERASMUS+ enrolment winter semester at Faculty of Fisheries and Protection of Waters, University of South Bohemia, Czech Republic

Track record

Original research articles

1. Nekvapil, F., Cardan, I., Cuibus, D., Farcau, C. Nanomolar concentration tracking of the carotenoid astaxanthin by surface-enhanced Raman spectroscopy. *Microchemical Journal*, 2026, 226: 118424; DOI: : [10.1016/j.microc.2026.118424](https://doi.org/10.1016/j.microc.2026.118424)
2. Nekvapil, F., Farcau, C. The role of focused laser plasmonics in shaping SERS spectra of molecules on nanostructured surface. *Nanoscale advances*, 2025, 7: 3008-3017; DOI: [10.1039/d4na00982g](https://doi.org/10.1039/d4na00982g).
3. Nekvapil, F., Stefan, M., Marica, I., Falamas, A. A novel observation upon the near-resonant Raman excitation of zinc oxide nanoparticles: Widening of the E2(high) band. *Journal of Raman Spectroscopy*, 2022, 51: 959-968; DOI: [10.1002/jrs.6323](https://doi.org/10.1002/jrs.6323).
4. Nekvapil, F., Stegarescu, A., Lung, I., Hirian, R., Cosma, D., Levei, E., Soran, M.-L. A Novel Nanoporous Adsorbent for Pesticides Obtained from Biogenic Calcium Carbonate Derived from Waste Crab Shells. *Nanomaterials*, 2023, 13: 3042; DOI: [10.3390/nano13233042](https://doi.org/10.3390/nano13233042).
5. Nekvapil, F., Bortnic, R-A., Leoştean, C., Barbu-Tudoran, L., Bunge, A. Characterization of the lattice transitions and impurities in manganese and zinc doped ferrite nanoparticles by Raman spectroscopy and x-ray diffraction (XRD). *Analytical Letters*, 2022. 56: 42-52; DOI: [10.1080/00032719.2022.2083145](https://doi.org/10.1080/00032719.2022.2083145).
6. Nekvapil, F., Ganea, I-V., Ciorîţă, A., Hirian, R., Tomšić, S., Martonos, I.M., Cintă Pinzaru, S. A New Biofertilizer Formulation with Enriched Nutrients Content from Wasted Algal Biomass Extracts Incorporated in Biogenic Powders. *Sustainability*, 2021, 13: 8777; DOI: [10.3390/su13168777](https://doi.org/10.3390/su13168777).
7. Nekvapil, F., Cintă Pinzaru, S., Barbu-Tudoran, L., Suci, M., Tamas, T., Chis, V. Color-specific porosity in double pigmented natural 3d-nanoarchitectures of blue crab shell. *Scientific Reports*, 2020, 10: 3019; DOI: [10.1038/s41598-020-60031-4](https://doi.org/10.1038/s41598-020-60031-4).
8. Nekvapil, F., Aluas, M., Barbu-Tudoran, L., Suci, M., Bortnic, R-A, Glamuzina, B., Cintă Pinzaru, S. From Blue Bioeconomy toward Circular Economy through High-Sensitivity Analytical Research

on Waste Blue Crab Shells. *ACS Sustainable Chemistry & Engineering*, 2019, 7: 16820-16827; DOI: [10.1021/acssuschemeng.9b04362](https://doi.org/10.1021/acssuschemeng.9b04362).

Patent

- Nekvapil, F., Soran, M-L., Stegarescu, A. Material adsorbant pentru poluanți obținut din deșeuri de cochilii de crab – metoda de obținere; Publicat in baza de date *Espacenet*, code RO138758A2 (URL: <https://worldwide.espacenet.com/patent/search/family/099821365/publication/RO138758A2?q=Fran%20Nekvapil>)

Internal INCDTIM Method file

- Determinarea distribuției ocupării siturilor în nanoparticulele de oxid de fier utilizând spectroscopia Raman; [INCDTIM registration nr. 2396/5.05.2025](#)

Narrative CV

A narrative summarizing which work has had the greatest importance and impact.

The interdisciplinary involvement of dr. Nekvapil began during his Master study cycle at the University of Dubrovnik. He studied “mariculture”, a science of cultivating aquatic organisms for human consumption and pharma industry, however, in the year 2015 he came into contact with his future Ph.D. mentor, Prof. Pinzaru from Babeş-Bolyai University (BBU), and became interested in high-precision spectroscopy analytical methods.

This occasional contact with world of physics developed into a collaboration where his Master thesis presented the applications of Raman spectroscopy in studies of marine organisms and chemical composition of their tissues. Further, he moved to the Faculty of Physics at the BBU, where he defended his doctoral thesis titled “Raman Spectroscopy Techniques and Applications to Carotenoids Research in Marine Living Resources and to the Blue Bioeconomy” in 2020.

Dr. Nekvapil has continued with the research of aquatic resources using high-precision analytical methods (Raman spectroscopy techniques, XRD, electron microscopy, etc.) and waste management. In 2018 he got employed at the Nat. Inst. For Research and Development of Isotopic and Molecular Technologies (INCDTIM), where he embarked onto another research direction concerning nanomaterials engineering, relying on his profound knowledge of spectroscopy analytical techniques. In 2021, he joined the [NanoPhotoniX](#) team at INCDTIM, whereafter he reduced the intensity of waste management research and instead focused more on investigations of fundamental spectroscopy processes and plasmonic mechanisms.

Current areas of work are concentrated around several topics, with varying degree of involvement:

Raman methodology development: The expertise of dr. Fran Nekvapil is applied Raman spectroscopy (RS). This includes various techniques, such as multi-laser Raman, Resonance Raman, Surface-enhanced Raman spectroscopy (SERS), and FT-Raman, closely supported by other analytical techniques depending on nature of the research subject, such as UV-Vis absorption, FT-IR (transmission and ATR modes), electron microscopy, XRD, magnetometry, N₂ physisorption (BET), etc. The PD does not routinely conduct non-spectroscopy measurements, however, he is capable to interpret and integrate the respective results into a comprehensive characterization of the sample at hand. Dr. Fran developed a sense for noting spectral patterns and in-depth analysis, deconvolution and interpretation of spectra.

SERS molecular tracking: dr. Nekvapil recently started working with self-assembled film-over-nanospheres (FoN) type plasmonic substrates. Through the use of AuFoN, he described the role of focused laser plasmonics

in shaping SERS spectra of molecules, coupled with FDTD modelling of near-field response of the nanostructured substrate. Further work includes testing the detection limits and interaction mechanisms at molecule-substrate interface. Earlier, dr. Nekvapil also worked extensively with colloidal silver nanoparticles (AgNPs), including their synthesis and low-concentration molecular detection.

Blue biotechnology: dr. Nekvapil has extensively worked, especially during his Ph.D. and the following years, on management and recycling of seafood waste and recovery of secondary materials. These include extraction of high-value natural antioxidants carotenoids, recovery of nanoporous biogenic calcium carbonate, and processing algal biomass into novel fertilizers.

Nanomaterials characterization: since employment at INCDTIM, dr. Nekvapil is in constant contact with ferrite and semiconductor (ZnO) nanoparticles. His primary role is to use multi-wavelength Raman microscopy techniques to establish structure-signal relationships and enable quality control of as-synthesized material. This topic extended beyond his immediate work environment and yielded nanomaterial-related collaborations with groups at Babeş-Bolyai University, Timișoara University of Medicine and Chemnitz University of Technology.

Microplastics in food systems: dr. Nekvapil also works part-time through the Faculty of Physics on an international project led by University of Geneva, partnered with University of Dubrovnik, targeting analytical and safety challenges posed by micro- and nano-plastics in seafood production chains.

Besides authored works, dr. Nekvapil was engaged into analysis and interpretation of spectra of various sample types by Raman spectroscopy techniques. Early on, he participated in a study of mineral and biological characterization of spring waters from Swiss Alps, then he identified minerals of palaeontological samples, assisted in characterization of advanced polymers, contributed to colleagues developing chitin extraction processes for Blue biotechnology, and participated in identification of polyene-type pigments in a range of biological matrices.

Dr. Nekvapil also mentored a Master student with aspects and tasks related to laboratory work regarding extraction and UV-Vis/Resonance Raman characterization of lycopene. As a senior member of the [Raman-AFM lab](#) at BBU in the group of prof. Pinzaru, Fran also assists with introductory lab sessions and desk research for newcomers and student exchange visitors.